Lesson 6

Recap

Printing a sentence using variables

1) f string-
print(f"Hello {name}, you are {age} years old.")

2) commas-
print("Hello", name, "you are", age, "years old.")

3) Concatenation (+)
print("Hello " + name + "!")
-cannot comcatenate ints with string

Recap

Comparison Operators

What they do: Compare two values and return **True or False**

Think of it like: Python is answering a yes or no question

Operator	Meaning	Example	Result
`==`	equal to	`5 == 5`	True
`!=`	not equal to	`5 != 3`	True
`>`	greater than	`10 > 3`	True
`<`	less than	`2 < 1`	False
`>=`	greater or equal	`5 >= 5`	True
`<=`	less or equal	`3 <= 2`	False



	`and`	`or`
True, True	✅ True	✅ True
True, False	❌ False	✅ True
False, False	❌ False	❌ False

Park Entry Checker

there should be 4 variables that ask the user for their name, age, height and if they have the wristband.

user is considered old enough if they are older than 12 and considered tall enough if taller than 5ft

To ride a ride called ultra sonic, one has to have a wristband, and either be old enough or tall enough.

Print if user is eligible to ride ultra sonic



Not

Just flip it

True becomes False
and False becomes
True

```
is_raining = False
bring_umbrella = not is_raining
```

and → both must be True
or → at least one must be True
not → flips True to False and False to True

We just built a program that prints True or False for whether someone can ride. But what if instead of printing True or False, we wanted it to print 'Welcome aboard!' or 'Sorry, you can't ride.

If Statements

let your code make decisions based on specific conditions

If it is raining, open an umbrella.

If you are 18 or older, you can vote




```
if condition:
    # code that runs if condition is True
```

Rewrite park entry checker program using if statements

What if the condition is false?

Else

else is what happens when the condition is False — it's the backup plan

else

```
if condition:
    # Code to run if the condition is True
else:
    # Code to run if the condition is False
```

else is optional; code does not crash if you don’t have an else

But it also won’t print out anything if condition is false

Even/Odd number checker

write a program that asks the user for a number and checks if it is even or odd. Should the print the correct statement.

hint: use an arithmetic operator we've learnt about

modulus

Practice Programs

example:

voting eligibility

```
age = int(input("Enter your age: "))

if age >= 18:
    print("You are eligible to vote!")
else:
    print("Sorry, you are too young to vote.")
```

Practice Programs

to do at home

1)Password checker

asks user to enter password, prints if it is correct(set your own correct password).

2) Positive or Negative number checker

user should enter number, print if it is neg or pos

3) A+ grade in Test or not

user should enter their grade, print if they got an A or not. (Grade above 90 equals A).